**“carpenters, blacksmiths, laborers, farm workers, and horse dealers”:** Ibid., p. 12.

**least expensive high school:** Free public secondary schools didn’t become universal in the UK until several years later (ibid., p. 13).

**“did the first year of my college physics”:** Ibid., p. 14.

**“Then the puzzles start”:** Ibid., p. 50.

**“bookkeeping operation”:** Mann and Crease 1988, p. 86.

**what was that information about?:** Bell 2004, p. 215.

**“we couldn’t get all that clear”:** Bernstein 1991, p. 51.

**“to epistemology, to philosophy, to humanity in general”:** Ibid., p. 52.

**“perfectly obscure”:** Ibid., pp. 52–53.

**“couldn’t interpret quantum mechanics”:** Ibid., p. 64.

**“a hole I wouldn’t get out of”:** Ibid., p. 53.

**“They did a fantastic somersault”:** Ibid., p. 66.

**“von Neumann’s unreasonable axiom”:** Ibid., p. 65.

**his talk about accelerators:** Ibid., p. 67.

**“if the lift had gotten stuck between floors”:** Mann and Crease 1988, p. 85.

**“not merely false but *foolish*!”:** Ibid., p. 88. Emphasis in original.

**so-called hidden variables:** Gleason’s proof actually didn’t mention hidden variables. Gleason was a mathematician, not a physicist, and his proof had to do with certain features of Hilbert space, the mathematical structure underlying quantum physics. But Jauch and his colleague Piron pointed out to Bell that Gleason’s proof had an obvious corollary that ruled out hidden variables—and that this corollary seemed far stronger than von Neumann’s proof, or their own.

**an unjustified assumption at the foundation:** Actually, Bell found two different assumptions: one used by von Neumann and Jauch, and one by Gleason. Gleason’s assumption is really what led Bell to contextuality. Von Neumann’s assumption is related to Gleason’s, but it’s more specific than Gleason’s, and rejecting it is easier. And it was von Neumann’s assumption that Grete Hermann correctly recognized as unwarranted in the 1930s and that Bell himself later characterized as “silly.”

**the same is true for evens and odds:** Technically, this isn’t how real roulette wheels are constructed (though it is how the roulette wheel in Figure 7.2 is constructed). There are ten red odds and eight red evens on